

Introduction to Version Control Systems

A Beginner's Guide to Git and GitHub

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What is Version Control?

- Tracks changes made to files over time.
- Maintains a history of modifications.
- Enables collaboration and rollback.

Types of Version Control Systems

DVCS

- Full repository on every machine.
- Examples: Git, Mercurial, Darcs.

CVCS

- Central server stores history.
- Examples: SVN, CVS.

Git and GitHub

- Git: Distributed version control system.
- Created by Linus Torvalds in 2005.
- GitHub: Cloud hosting platform for Git repositories.

Installing Git

Windows:

- 1 Download Git from git-scm.com.
- 2 Run installer and complete setup.
- 3 Verify:

```
git --version
```

Linux:

```
sudo apt update  
sudo apt install git  
git --version
```

Configure Git

```
git config --global user.name "Your Name"  
git config --global user.email "you@example.com"
```

Creating a Repository

```
git init  
git status  
git log
```

Working Directory → Staging Area → Commit History

Add and Commit

```
git add .  
git commit -m "Initial commit"
```

```
winget install GitHub.cli  
gh auth login
```

Create and Push Repository

```
gh repo create PROJECT_NAME —public —source=. —push
```

Clone a Repository

```
git clone REPOSITORY_URL
```

Connect Existing Repository

```
git remote add origin REPOSITORY_URL
git add .
git commit -m "Initial Commit"
git push --set-upstream origin master
```

Summary

- Git tracks project history.
- GitHub hosts repositories online.
- Core commands: init, add, commit, push, clone.